

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Preliminary Examination Paper 2 Answer Book

NAME: Memo

CLASS: 12 _____ TEACHER: _____

	Mark				
Question	H	T	U	Marker	HOD
1					
2					
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TOTAL					
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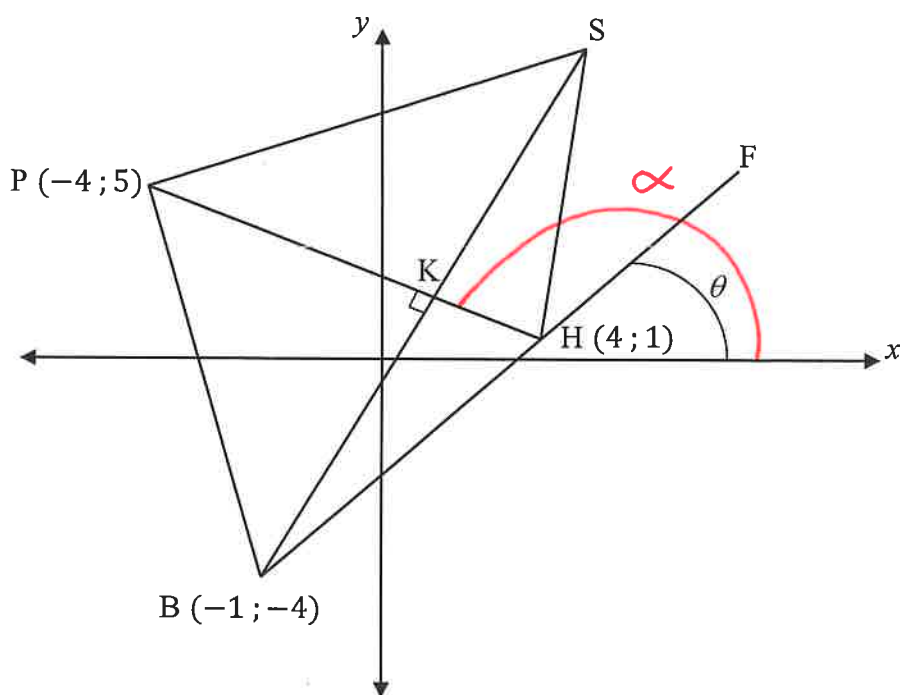
QUESTION 1

Player	Age in years (x)	Number of caps (y)
Canan Moodie	20 <i>min</i>	3
RG Snyman	28	23
Jesse Kriel	29 <i>Q₁</i>	57
Damian De Allende	31	61
Eben Etzebeth	<u>31</u> <i>Q₂</i>	112
Pieter-Steph Du Toit	31	61
Siya Kolisi	32	74
Frans Malherbe	32 <i>Q₃</i>	61
Trevor Nyakane	34	57
Willie Le Roux	34 <i>max</i>	87

	Solution	Marks
1.1.1	$\bar{y} = 59,6$ <i>✓A</i>	(2)
	$\sigma = 28,75$ <i>✓A</i>	
1.1.2	$\bar{y} - \sigma = 30,85$ <i>✓m</i>	(2)
	$\therefore 2$ Players <i>✓CA</i>	
1.2.1	<i>✓min, max</i> 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34	(4)
1.2.2	Negatively skewed <i>✓A</i>	(1)

	Solution	Marks
1.3.1	$\hat{y} = a + bx$ ✓ ^m	
	$\hat{y} = -109,77 + 5,61x$ ✓ ^A ✓ ^A	(3)
1.3.2	$r = 0,75$ ✓ ^A	
1.3.3	Moderate ✓ positive ✓ linear	
	Alt: the older the player, the more caps he will have	
1.3.4	$\hat{y} = -109,77 + 5,61(33)$ ✓ ^m	
	$\hat{y} = 75,36$ ✓	
1.3.5	Reliable ✓	
	33 years lies within age range ✓	
	Alt: correlation is moderate	(2)
		[19]

QUESTION 2

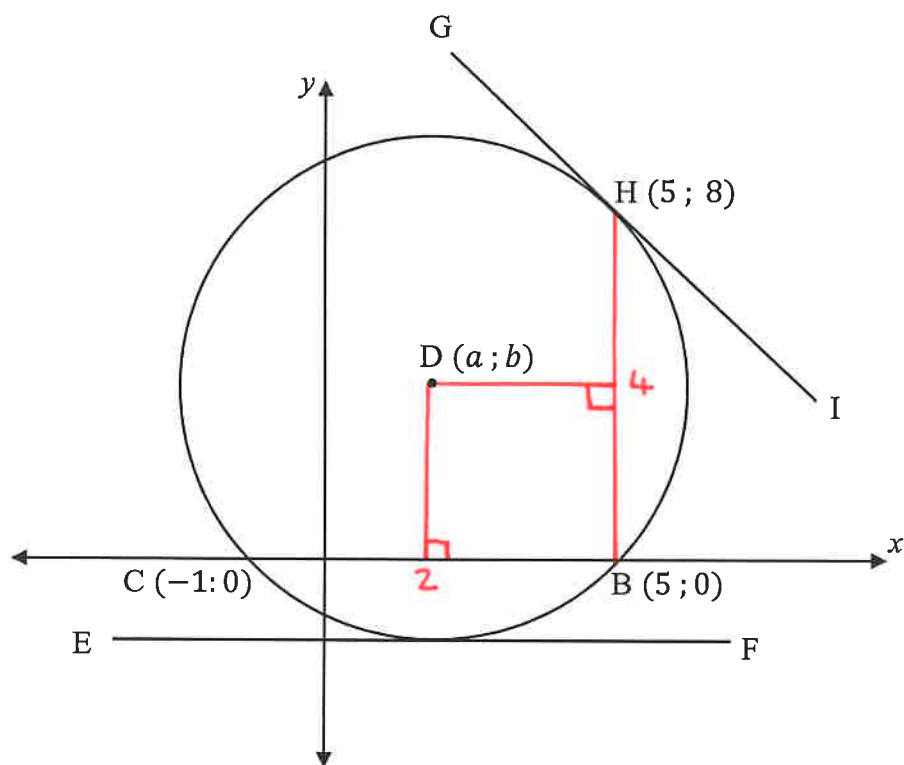


	Solution	Marks
2.1	$m_{PH} = \frac{5-1}{-4-4} = -\frac{1}{2}$	(2)
2.2	$m_{BS} = 2$ $y+4 = 2(x+1)$ $y = 2x - 2$	
2.3	$PH: y-1 = -\frac{1}{2}(x-4)$ $y = -\frac{1}{2}x + 3$ $-\frac{1}{2}x + 3 = 2x - 2$ $-x + 6 = 4x - 4$ $5x = 10$ $\therefore x = 2$ $\therefore y = 2(2) - 2 = 2$ $K(2; 2)$	(4)

$$y = 2$$

	Solution	Marks
2.4	$m_{BH} = \frac{1+4}{4+1} = 1 \checkmark A$ $\tan \theta = 1$ $\therefore \theta = 45^\circ \checkmark$ $\checkmark m (\tan \angle = m)$ $\tan \alpha = -\frac{1}{2}$ $\alpha = 153,43^\circ \checkmark$ $\therefore \hat{PHF} = 153,43 - 45^\circ$ $= 108,43^\circ \checkmark CA$	(5)
2.5	$BK = \sqrt{(2+1)^2 + (2+4)^2} = 3\sqrt{5} \checkmark A$ $PH = \sqrt{(-4-4)^2 + (5-1)^2} = 4\sqrt{5} \checkmark A$ $\text{Area of } \triangle PBH = \frac{1}{2} (3\sqrt{5})(4\sqrt{5}) \checkmark m$ $= 30 \text{ units}^2 \checkmark CA$	(4)
2.6	$M_{BH} \left(\frac{-1+4}{2}, \frac{-4+1}{2} \right)$ $= \left(\frac{3}{2}, -\frac{3}{2} \right) \checkmark A$ $M_{BH} = M_{PA} \quad \checkmark R \quad \text{Diags of parm}$ $\frac{x_A - 4}{2} = \frac{3}{2} \quad \frac{y_A + 5}{2} = -\frac{3}{2}$ $\therefore A(7; -8) \checkmark CA$	(4)
		[22]

QUESTION 3



	Solution	Marks
3.1	a is midpoint of BC (line from $\checkmark$) b is midpoint of BH (centre $\perp$ chord) <i>*Can show on sketch</i> $\therefore a = 2$ $b = 4$	(2)
3.2	$(x-2)^2 + (y-4)^2 = r^2$ $(5-2)^2 + (0-4)^2 = r^2$ $\therefore r^2 = 25$ $(x-2)^2 + (y-4)^2 = 25$ $\checkmark$	(3)
3.3	$y = -1$ $\checkmark$ Horizontal line $y = k$ $\checkmark -1$	(2)

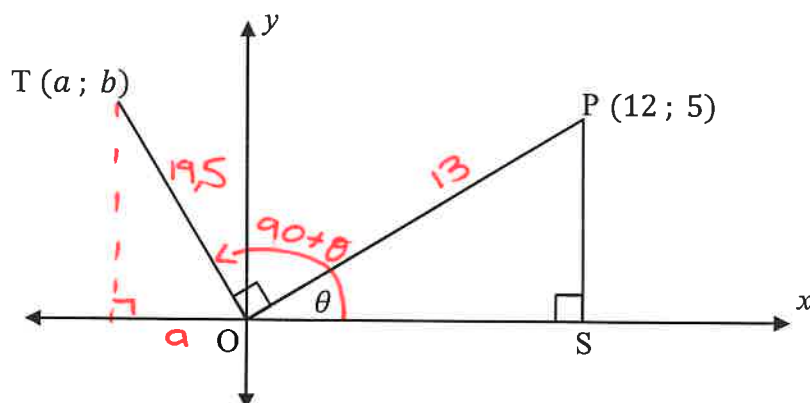
→ Only 2 gradients required.

	Solution	Marks
3.4	$m_{CD} = \frac{4-0}{2+1} = \frac{4}{3}$ $m_{DH} = \frac{8-4}{5-2} = \frac{4}{3}$ $m_{CH} = \frac{8-0}{5+1} = \frac{4}{3}$ $m_{CD} = m_{DH} = m_{CH} \quad \checkmark A$ $\therefore C, D$ and H are collinear	(3)
3.5	$m_{rad} = \frac{4}{3}$ $m_{tan} = -\frac{3}{4} \quad \checkmark R \quad rad \perp tan$ $y-8 = -\frac{3}{4}(x-5) \quad \checkmark m$ $y = -\frac{3}{4}x + \frac{47}{4} \quad \checkmark CA$	(3)

	Solution	Marks
3.6	$r_2 - r_1 < DN < r_2 + r_1$	
	$5 - 4 < DN < 5 + 4$	
	$1 < DN < 9$ ✓m	
	$1 < p - 2 < 9$ ✓ or $1 < 2 - p < 9$ ✓	
	$3 < p < 11$ ✓ or $-1 < -p < 7$	
	$-7 < p < +1$ ✓	
		(5)
		[18]

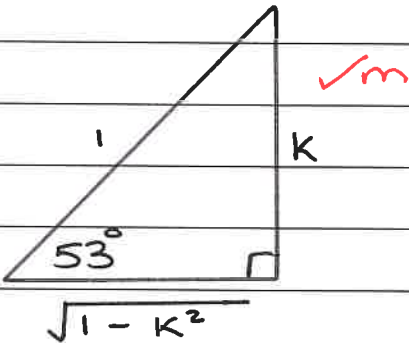
QUESTION 4

4.1

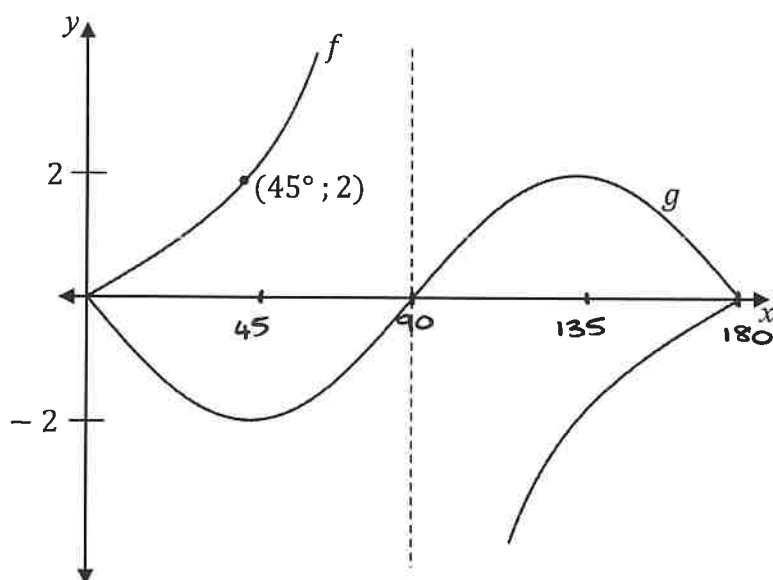


	Solution	Marks
4.1.1	$\tan \theta = \frac{5}{12}$ ✓A	
		(1)

	Solution	Marks
4.1.2	$\sin \theta = \frac{5}{13}$ $r^2 = 12^2 + 5^2$ $r^2 = 169$ $r = 13$	(2)
4.1.3	$\cos(90 + \theta) = \frac{a}{19,5}$ $-\sin \theta = \frac{a}{19,5}$ $-\frac{5}{13} = \frac{a}{19,5} \quad \therefore a = -\frac{15}{2} \approx -7,5$	(3)
4.2	$= \frac{-\cos x \cdot (-\tan x) \cdot \cos^2 x}{\sin x} - \sin^2 x$ $= \cos^2 x - \sin^2 x$ $= \cos 2x$	(6)
4.3	$\cos 105$ $= -\cos 75$ $= -\cos(45 + 30)$ $= -\cos 45 \cos 30 + \sin 45 \sin 30$ $= -\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2}$ $= \frac{-\sqrt{6} + \sqrt{2}}{4}$	(4)

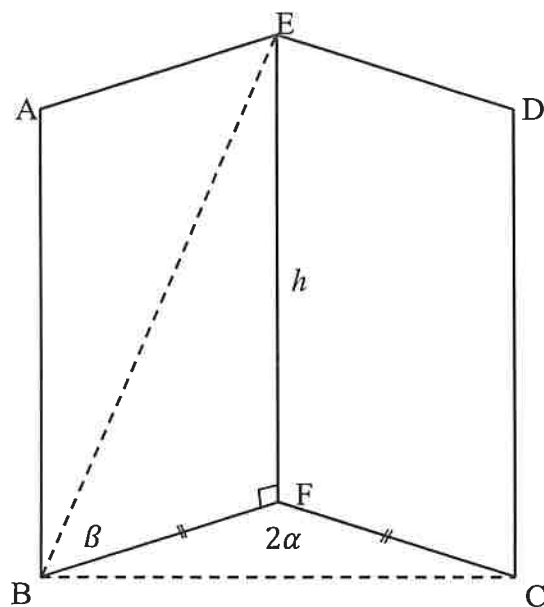
	Solution	Marks
4.4	$= \sin(31+22)$ $\sin 53 = k \checkmark$ $\tan 53 = \frac{k}{\sqrt{1-k^2}} \checkmark \text{CA}$ 	(4)
4.5	$\text{LHS} = \frac{\sin(2\theta + \theta)}{\sin \theta} \checkmark$ $= \frac{\sin 2\theta \cos \theta + \cos 2\theta \sin \theta}{\sin \theta} \checkmark \text{compound}$ $= \frac{2 \sin \theta \cos^2 \theta + \cos 2\theta \sin \theta}{\sin \theta} \checkmark \text{sin } 2\theta$ $= \frac{\sin \theta (2 \cos^2 \theta + \cos 2\theta - 1)}{\sin \theta} \checkmark \text{cos } 2\theta$ $= 4 \cos^2 \theta - 1 = \text{RHS}$ $\therefore \text{LHS} = \text{RHS}$	(5)
4.6	$2 \cos^2 x = 3 \sin x$ $2(1 - \sin^2 x) = 3 \sin x \checkmark$ $2 - 2 \sin^2 x = 3 \sin x$ $2 \sin^2 x + 3 \sin x - 2 = 0 \checkmark A$ $(2 \sin x - 1)(\sin x + 2) = 0$ $\sin x = \frac{1}{2} \checkmark \text{CA both} \quad \text{or} \quad \sin x = -2 \checkmark \text{reject}$ No Solution $\therefore x = 30^\circ + k \cdot 360^\circ ; k \in \mathbb{Z} \checkmark \text{CA both}$ $\text{or } x = 150^\circ + k \cdot 360^\circ ; k \in \mathbb{Z}$	(5)
		[30]

QUESTION 5



	Solution	Marks
5.1	$a = 2$ ✓ $b = -2$ ✓ $k = 2$ ✓	(3)
5.2	$0 \leq x < 90^\circ$; $x = 180^\circ$ ✓ ✓ Inequalities ✓ Values	(3)
5.3	$h(x) = -2 \tan x + 2$ ✓	(2)
5.4	$x = 90^\circ$ ✓	(1)
5.5	$y = 2 \cos 2(x + 45)$ ✓ Alts: $y = 2 \cos (2x + 90)$ $y = -2 \cos (2x - 90)$ $y = -2 \cos 2(x - 45)$	(1)
		[10]

QUESTION 6

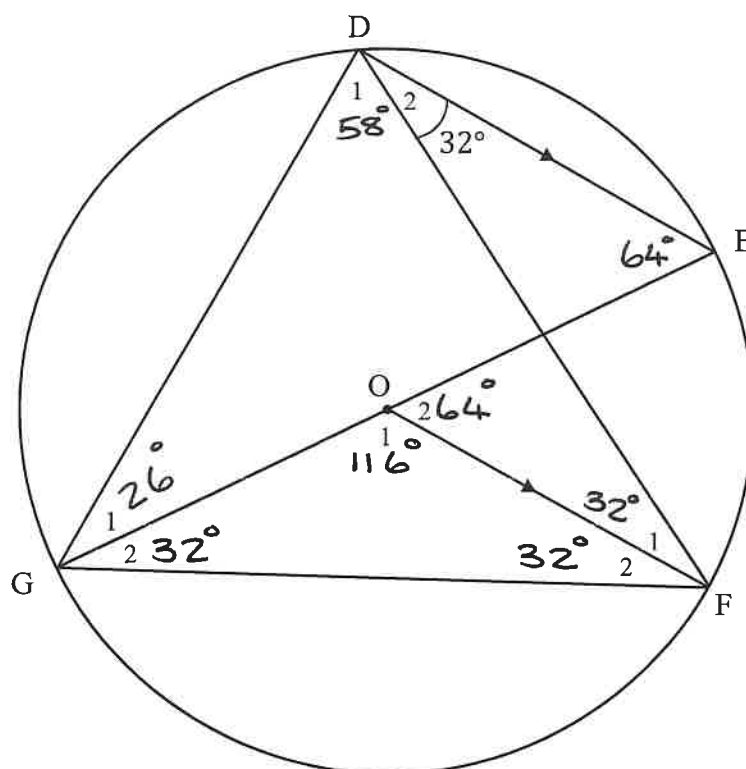


	Solution	Marks
6.1	In $\triangle BEF$: $\tan \beta = \frac{h}{BF}$ $\therefore BF = \frac{h}{\tan \beta}$ ✓A In $\triangle BCF$: $\hat{FBC} = \hat{FCB}$ ✓R $\angle$'s opp = sides $\hat{FBC} = \hat{FCB} = 90 - \alpha$ ✓R $\angle$'s in a $\triangle$ $\frac{BC}{\sin \hat{BFC}} = \frac{BF}{\sin \hat{BCF}}$ ✓sin rule $\frac{BC}{\sin 2\alpha} = \frac{BF}{\sin (90 - \alpha)}$ $BC = BF \cdot \frac{2 \sin \alpha \cos \alpha}{\cos \alpha}$ ✓ $= \frac{h}{\tan \beta} \cdot \frac{2 \sin \alpha \cos \alpha}{\cos \alpha}$ ✓SUBS $= \frac{2h \cdot \sin \alpha}{\tan \beta}$	(7)

$\tan \beta$

	Solution	Marks
6.2	$25 = \frac{2h \cdot \sin 50}{\tan 55}$ ✓subs	(2)
	$\therefore h = 23,30\text{cm}$ ✓ca	
		[9]

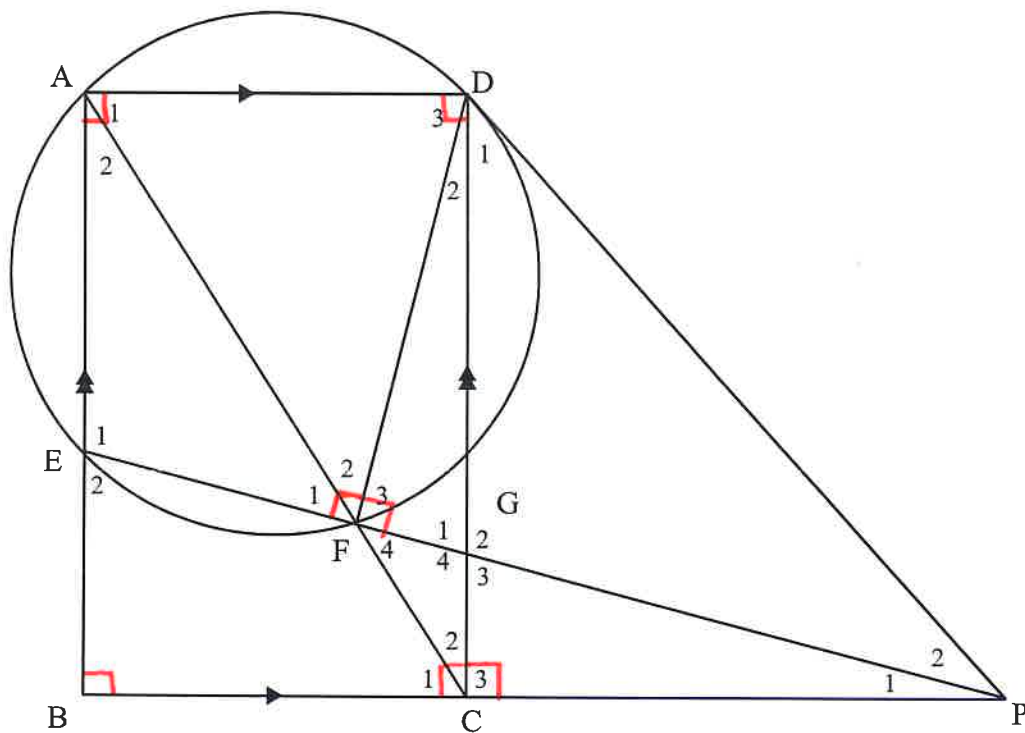
QUESTION 7



	Solution	Marks
7.1	$\hat{GDE} = 90^\circ$ ✓ $\hat{D}_1 = 58^\circ$ ✓	(2)
	✓ _R $\angle$ in semi $\odot$	
7.2	$\hat{O}_1 = 116^\circ$ ✓	(2)
	$\angle$ at centre = $2 \times \angle$ at circ ✓	
7.3	$\hat{G}_2 = \hat{F}_2$ ✓ $\hat{F}_2 = 32^\circ$ ✓	(2)
	✓ _R $\angle$'s opp = radii	
	✓ _R $\angle$'s in a Δ	

	Solution	Marks
7.4	$\hat{O}_2 = 64^\circ$ ✓ $\angle$ at centre = $2 \times \angle$ at circ Alt: Ext $\angle$ of Δ $\therefore \hat{E} = 64^\circ$ ✓ Alt $\angle$ s ; $DE \parallel OF$ ✓	(3)
		[9]

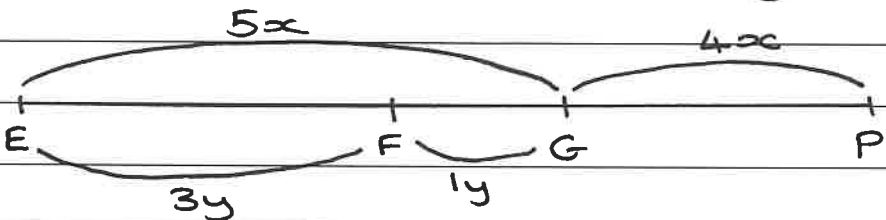
QUESTION 8



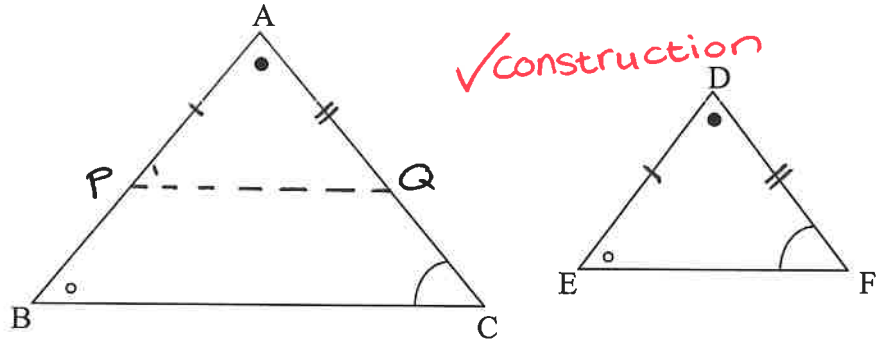
	Solution	Marks
8.1	$\hat{DAB} = 90^\circ$ S/R Int $\angle$ of rectangle	
	$\hat{EFD} = 90^\circ$ S/R Opp $\angle$'s of cyclic quad	
	$\therefore EP \perp FD$	

(2)

	Solution	Marks
8.2	$\hat{C}_3 = 90^\circ$ $\angle$'s on a line	
	$\hat{F}_3 = 90^\circ$ Ext $\angle$ of cyclic quad	
	$\therefore \hat{C}_3 = \hat{F}_3$ ✓ Only award if reasoning is correct.	
	DFCP is a cyclic quad Converse $\angle$'s in same segment ✓ _R	
		(2)
8.3	$\hat{A}_1 = \hat{C}_1$ $\checkmark_R$ Alt $\angle$'s ; $AD \parallel BC$	
	$\hat{C}_1 = \hat{PDF}$ $\checkmark_R$ Ext $\angle$ of cyclic quad	
	$\therefore \hat{A}_1 = \hat{PDF}$ ✓	
	$\therefore DP$ is a tangent Converse tan chord thrm ✓ _R	
		(4)

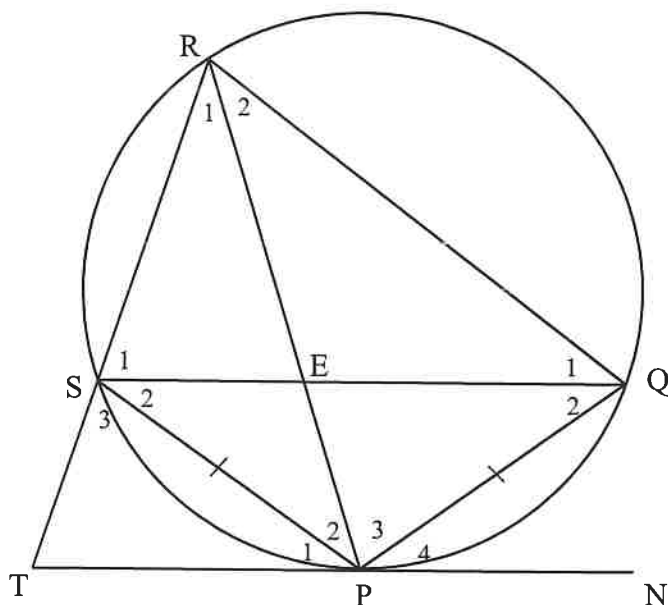
	Solution	Marks
8.4	$\frac{BC}{CP} = \frac{5}{4} \qquad \frac{EF}{FG} = \frac{3}{1}$ $\frac{BC}{CP} = \frac{EG}{GP} = \frac{5}{4} \quad \text{S/R line } \parallel \text{ side of } \Delta$ Let $EG = 5x$, $GP = 4x$, $EF = 3y$, $FG = 1y$  $EG : 5x = 4y$ $\therefore y = \frac{5x}{4}$ $EP : 9x = 36$ $x = 4 \text{ units } \checkmark$ $\therefore y = \frac{5}{4} (4) = 5$ $FG = 1y$ $FG = 5 \text{ units } \checkmark \text{CA}$	
		(4)
		[12]

QUESTION 9

	Solution	Marks
9.1	 In $\triangle APQ$ and $\triangle DEF$ $\hat{A} = \hat{D}$ Given $AP = DE$ Construction $AQ = DF$ Construction $\therefore \triangle APQ \equiv \triangle DEF$ SAS ✓ $\equiv \Delta$'s proof $\therefore \hat{P}_1 = \hat{E}$ ✓ $\equiv \Delta$'s $\hat{E} = \hat{B}$ Given $\therefore \hat{P}_1 = \hat{B}$ $\therefore PQ \parallel BC$ ✓ Corr $\angle$'s = $\therefore \frac{AB}{AP} = \frac{AC}{AQ}$ ✓ line $\parallel$ side of Δ But $AP = DE$ and $AQ = DF$ $\therefore \frac{AB}{DE} = \frac{AC}{DF}$ ✓ Similarly $\frac{AB}{DE} = \frac{BC}{EF}$ $\therefore \frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$	

(6)

9.2



	Solution	Marks
9.2.1	$\hat{P}_1 = \hat{Q}_2$ S/R tan chord thrm $\hat{Q}_2 = \hat{S}_2$ S/R $\angle$'s opp = sides $\therefore \hat{P}_1 = \hat{S}_2$ ✓ $\therefore TN \parallel SQ$ Alt $\angle$'s = ✓R	(4)
9.2.2	$\hat{T}$ is common ✓ $\hat{P}_1 = \hat{R}_1$ S/R tan chord thrm $\therefore \hat{S}_3 = \hat{T}_1$ S/R 3 rd $\angle$ in Δ $\therefore \Delta TSP \parallel \Delta TPR$ <<< ✓R	(4)

	Solution	Marks
9.2.3	$\hat{R}_1 = \hat{R}_2$ S/R Equal chords $\rightarrow$ Equal $\angle$'s $\hat{R}_1 = \hat{P}_1$ Proven $\therefore \hat{R}_2 = \hat{P}_1$ $\hat{S}_3 = \hat{RQP}$ S/R Ext $\angle$ of cyclic quad $\therefore \hat{STP} = \hat{P}_3$ S/R 3 rd $\angle$ of Δ $\therefore \Delta TSP \parallel \Delta PQR$ $\angle \angle \angle$ $\therefore \frac{TS}{PQ} = \frac{SP}{QR} = \frac{TP}{PR}$ S/R $\parallel \Delta$'s But $SP = QP$ $\therefore \frac{TS}{SP} = \frac{SP}{QR}$ ✓ $\therefore TS \cdot QR = SP^2$	
		(5)

	Solution	Marks
9.2.4	$\triangle TSP \parallel \triangle TPR$ Proven	
	$\triangle TSP \parallel \triangle PQR$ Proven	
	$\therefore \triangle TPR \parallel \triangle PQR$ ✓	
	$\therefore \frac{PR}{QR} = \frac{TR}{PR}$ ✓ $\parallel \triangle$'s	
	$\therefore PR^2 = TR \cdot QR$	(2)
		[21]